DRIVER DROWSINESS AND DISTRACTION DETECTION SYSTEM USING COMPUTER VISION

A PROJECT REPORT

Submitted by

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in partial fulfillment for the award of the

degree of

BACHELOR OF ENGINEERING

IN

MECHANICAL ENGINEERING

KNOWLEDGE INSTITUTE OF TECHNOLOGY, SALEM

ANNA UNIVERSITY:: CHENNAI 600 025

JULY 2026

BONAFIDE CERTIFICATE

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ABSTRACT

Road accidents caused by driver drowsiness and distraction are among the leading causes of injuries and fatalities worldwide. This project, titled "Driver Drowsiness and Distraction Detection System using Computer Vision," presents a real-time software-based driver monitoring system developed using Python in Visual Studio Code. The system continuously monitors the driver's facial features through an HD webcam and applies computer vision and artificial intelligence techniques to identify signs of fatigue and distraction. OpenCV is used for image processing, while MediaPipe Face Mesh detects facial landmarks required to calculate the Eye Aspect Ratio (EAR) for drowsiness detection and the Mouth Aspect Ratio (MAR) for yawn detection. The system also analyzes the driver's head position to identify distraction and employs the YOLOv8 object detection model to detect mobile phone usage while driving. Based on these analyses, the driver's condition is classified as Alert, Drowsy, Yawning, or Distracted, and the corresponding status is displayed in real time. Whenever unsafe driving behavior is detected, the system generates visual warnings, activates a software-based buzzer alert, automatically captures screenshots, and records the event details with the current date and time in a CSV log file for future analysis. The proposed system is a low-cost, non-contact, and efficient solution that demonstrates the practical application of computer vision and artificial intelligence for enhancing driver safety and reducing road accidents caused by fatigue and distraction.

Keywords: Driver Drowsiness Detection, Driver Distraction Detection, Computer Vision, Artificial Intelligence, OpenCV, MediaPipe Face Mesh, YOLOv8, Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), Driver Monitoring System, Real-Time Detection.

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1.SYMBOLS:

Symbol	Description
$<$	Less than (used for threshold comparison)
$>$	Greater than (used for threshold comparison)
$\leftarrow$	Assignment operator used in pseudocode
$\neq$	Not equal to
()	Parentheses used for function calls and parameter grouping

2. ABBREVIATIONS:

Abbreviation	Expansion
AI	Artificial Intelligence
CV	Computer Vision
CNN	Convolutional Neural Network
CSV	Comma-Separated Values
EAR	Eye Aspect Ratio
MAR	Mouth Aspect Ratio
FPS	Frames Per Second
HD	High Definition
IDE	Integrated Development Environment
IoT	Internet of Things
ML	Machine Learning
RAM	Random Access Memory
RGB	Red, Green and Blue
SSD	Solid State Drive
USB	Universal Serial Bus
YOLO	You Only Look Once
YOLOv8	You Only Look Once Version 8

3.NOMENCLATURE:

Nomenclature	Description
Driver State	Current condition of the driver (Alert, Drowsy, Yawning or Distracted)
Eye Aspect Ratio (EAR)	Mathematical ratio used to determine eye closure for drowsiness detection
Mouth Aspect Ratio (MAR)	Mathematical ratio used to detect yawning
Face Mesh	Facial landmark detection model used to locate facial key points
Facial Landmarks	Key points extracted from the driver's face for feature analysis
Head Position	Orientation of the driver's head used to determine distraction
Mobile Phone Detection	Identification of mobile phone usage using the YOLOv8 model
Real-Time Monitoring	Continuous processing of live webcam video without noticeable delay
Screenshot Capture	Automatic saving of unsafe driving events as image files
Event Logging	Recording driver events into a CSV file with date and time
Webcam	Image acquisition device used to capture the driver's face
Visual Warning	On-screen alert displayed when unsafe driving behavior is detected
Buzzer Alert	Audible warning generated when drowsiness or distraction is detected
MediaPipe Face Mesh	Framework used for real-time facial landmark detection
YOLOv8 Object Detection	Deep learning model used to detect mobile phone usage

CHAPTER 1
INTRODUCTION

1. Introduction

Road transportation is one of the most widely used modes of transportation worldwide. However, driver drowsiness and distraction are among the major causes of road accidents, resulting in significant loss of life and property every year. Fatigue caused by long driving hours, lack of sleep, and the use of mobile phones while driving can reduce a driver's attention and reaction time, increasing the risk of accidents.

Advancements in Computer Vision (CV) and Artificial Intelligence (AI) have enabled the development of intelligent driver monitoring systems that can continuously observe a driver's behavior in real time. These systems help identify signs of fatigue and distraction before they lead to dangerous situations, thereby improving road safety.

This project, titled "Driver Drowsiness and Distraction Detection System using Computer Vision," is a software-based solution developed using Python in Visual Studio Code. The system captures live video through a webcam and analyzes the driver's facial features using MediaPipe Face Mesh and OpenCV. It detects drowsiness by calculating the Eye Aspect Ratio (EAR), yawning through the Mouth Aspect Ratio (MAR), monitors head position to identify distraction, and uses YOLOv8 to detect mobile phone usage while driving.

Whenever unsafe driving behavior is identified, the system generates visual warnings and a software-based buzzer alert, captures screenshots of the event, and stores the event details along with the current date and time in a CSV log file. The proposed system is a low-cost, non-contact, and real-time solution that demonstrates the practical application of computer vision and deep learning for enhancing driver safety and reducing accidents caused by fatigue and distraction.

1.2 Methodology

The proposed Driver Drowsiness and Distraction Detection System follows a computer vision-based methodology for continuously monitoring the driver's condition. A webcam captures live video frames, which are processed using OpenCV. MediaPipe Face Mesh detects facial landmarks required to calculate the Eye Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR) for identifying drowsiness and yawning, respectively. The head position is also analyzed to determine whether the driver is looking forward or is distracted.

For detecting mobile phone usage, the system utilizes the YOLOv8 object detection model, which identifies the presence of a mobile phone in the camera frame. Based on the outputs from facial landmark analysis and object detection, the system classifies the driver's state as Alert, Drowsy, Yawning, or Distracted. When an unsafe condition is detected, the system immediately displays visual warnings on the monitoring dashboard, activates a software-generated buzzer alert, captures a screenshot of the event, and records the event details, including the current date and time, in a CSV log file. This methodology enables continuous, real-time monitoring of driver behavior without requiring any additional hardware prototype, making the system cost-effective and suitable for future integration into intelligent vehicle safety systems.

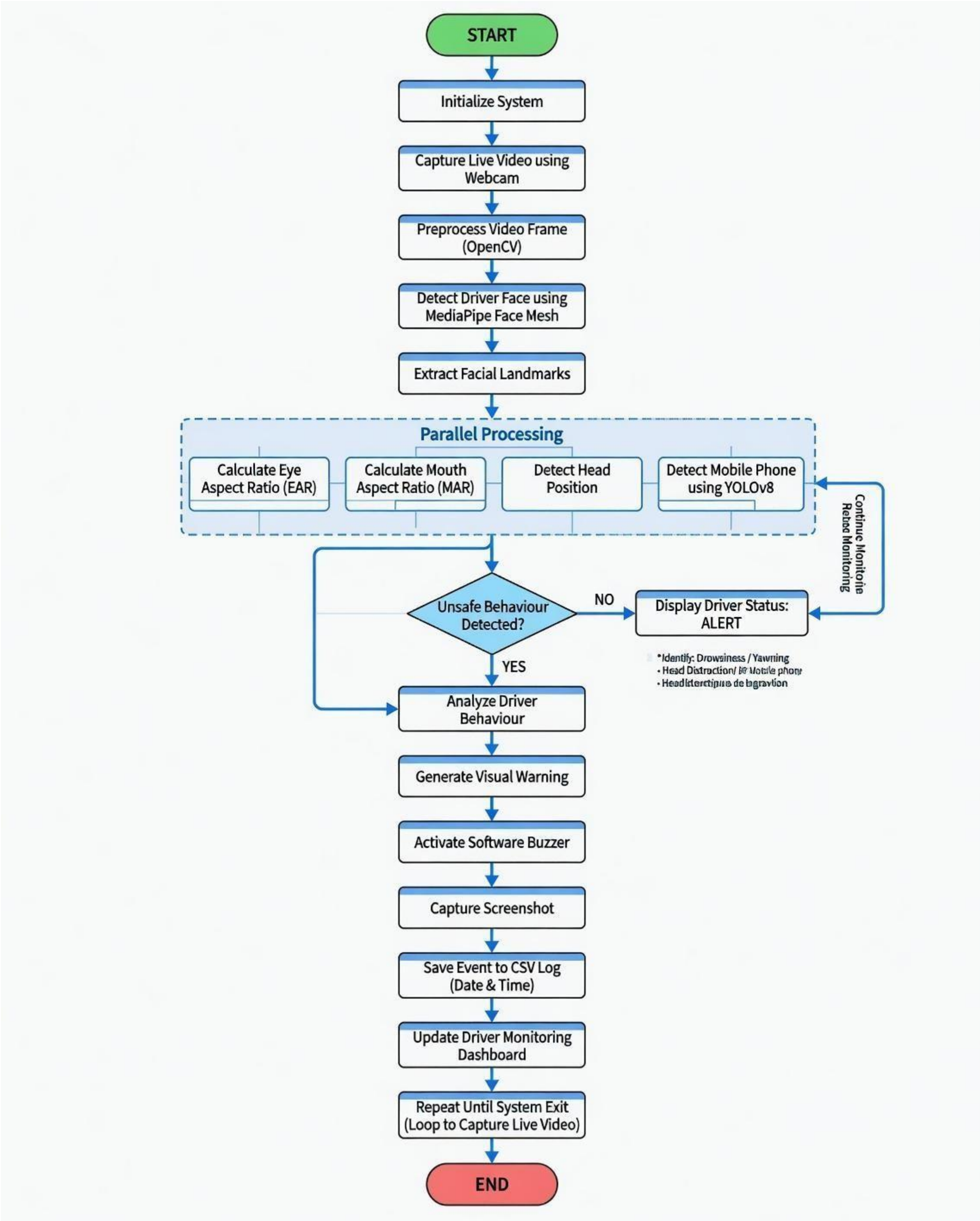

Figure 2.1: Working Methodology of the Proposed Driver Drowsiness and Distraction Detection System

1.3 Objectives

The primary objective of this project is to develop an intelligent Driver Drowsiness and Distraction Detection System using Computer Vision that continuously monitors the driver's behavior in real time and helps improve road safety by detecting unsafe driving conditions.

The specific objectives of the project are:

- To develop a real-time driver monitoring system using Computer Vision and Artificial Intelligence techniques.
- To detect driver drowsiness by calculating the Eye Aspect Ratio (EAR) using facial landmarks.
- To identify yawning by calculating the Mouth Aspect Ratio (MAR) as an indicator of driver fatigue.
- To monitor the driver's head position for detecting distraction from the road.
- To detect mobile phone usage while driving using the YOLOv8 object detection model.
- To provide immediate visual warnings and a software-generated buzzer alert whenever unsafe driving behavior is detected.
- To automatically capture screenshots of critical events for future analysis.
- To maintain a CSV log containing the event details along with the current date and time.
- To develop a low-cost, software-based, and real-time driver monitoring system that can contribute to reducing road accidents caused by fatigue and distraction.

1.4 Scope of the Project

The scope of the proposed Driver Drowsiness and Distraction Detection System using Computer Vision is focused on enhancing driver safety through continuous real-time monitoring of driver behavior using a webcam and computer vision techniques. The system operates without requiring any wearable sensors or specialized hardware, making it a cost-effective and non-invasive solution.

The project includes the following scope:

- Real-time video capture using a standard webcam.
- Driver face detection and facial landmark extraction using MediaPipe Face Mesh.
- Detection of driver drowsiness through Eye Aspect Ratio (EAR) analysis.
- Detection of yawning using Mouth Aspect Ratio (MAR).
- Head position monitoring to identify driver distraction.
- Mobile phone detection using the YOLOv8 deep learning model.
- Display of driver status through a real-time monitoring dashboard.
- Generation of visual warnings and software-based buzzer alerts during unsafe driving conditions.
- Automatic screenshot capture and event logging with the current date and time for documentation and analysis.
- Development and testing of the system using Python, OpenCV, MediaPipe, and YOLOv8 in the Visual Studio Code environment.

CHAPTER 2
LITRATURE SURVEY

Sankar et al. (2025), in *"Real-time Driver Drowsiness Detection using Transformer Architectures: A Novel Deep Learning Approach,"* proposed a transformer-based deep learning model for detecting driver drowsiness from facial features. The study demonstrated that transformer architectures improve feature extraction and classification accuracy compared to conventional convolutional neural networks, enabling reliable real-time fatigue detection.

Kumar et al. (2025), in *"Real-Time Driver Distraction Detection Using Fast R-CNN Algorithm,"* developed a distraction detection system using the Fast R-CNN object detection algorithm. Their work focused on identifying distracting activities such as mobile phone usage and demonstrated that deep learning-based object detection can effectively recognize unsafe driver behavior in real time.

Patel et al. (2026), in *"Intelligent Driver Monitoring Systems: A Survey of Drowsiness Detection Technologies for Road Safety,"* presented a comprehensive review of existing driver monitoring techniques. The survey discussed traditional image processing, machine learning, and deep learning approaches, highlighting the advantages and limitations of each method while emphasizing the need for integrated real-time monitoring systems.

Zhang et al. (2025), in *"Driver Distraction Detection Using a Multi-stream Deep Fusion Network,"* introduced a multi-stream deep learning framework that combines facial features, head pose, and contextual information for distraction detection. The proposed fusion network improved detection performance by integrating multiple visual cues instead of relying on a single feature.

Reddy et al. (2025), in *"Real-Time Driver Drowsiness Detection Using CNN-Based Eye Blink Analysis for Accident Prevention,"* proposed a convolutional neural network (CNN) model for analyzing eye-blink patterns to detect driver fatigue. The study achieved high detection accuracy and demonstrated that continuous eye monitoring is an effective approach for preventing fatigue-related accidents.

Sharma et al. (2025), in *"Driver Drowsiness Detection in Indian Vehicles with Face Obstructions,"* investigated the challenges of detecting drowsiness under practical driving conditions, including face masks, spectacles, and partial facial occlusions. Their work improved the robustness of driver monitoring systems in real-world environments where facial visibility is often limited.

The literature survey indicates that existing studies primarily focus on individual aspects such as drowsiness detection, distraction detection, or eye-blink analysis. However, limited research integrates multiple driver safety features into a single real-time system. Motivated by these observations, the proposed Driver Drowsiness and Distraction Detection System using Computer Vision combines Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), head position analysis, and YOLOv8-based mobile phone detection to provide a comprehensive, real-time driver monitoring solution. Additionally, the system generates visual alerts, captures event screenshots, and maintains CSV logs, making it a practical and cost-effective approach for enhancing road safety.

CHAPTER 3

PROPOSED SYSTEM

3.1 Proposed System

The proposed Driver Drowsiness and Distraction Detection System using Computer Vision is developed to continuously monitor the driver's condition and detect fatigue and distraction in real time. The system utilizes a webcam to capture live video frames and applies computer vision and deep learning techniques to analyze the driver's facial features and activities.

The proposed system consists of four primary detection modules: Eye Detection, Yawn Detection, Head Position Detection, and Mobile Phone Detection. Eye Aspect Ratio (EAR) is used to detect prolonged eye closure, while Mouth Aspect Ratio (MAR) is used to identify yawning. Head position estimation determines whether the driver is looking away from the road, and the YOLOv8 object detection model detects mobile phone usage.

The outputs from these modules are integrated to determine the driver's current state as Alert, Drowsy, Yawning, or Distracted. When unsafe behavior is detected, the system immediately displays warning messages, activates a software buzzer, captures a screenshot of the event, and records the details with the current date and time in a CSV log file. A real-time monitoring dashboard provides continuous visual feedback, enabling effective monitoring of the driver's condition.

The proposed system is implemented entirely in software using Python, OpenCV, MediaPipe Face Mesh, and YOLOv8, making it a cost-effective, non-contact, and real-time solution for improving road safety.

3.2 System Architecture

The proposed Driver Drowsiness and Distraction Detection System using Computer Vision is designed as a modular software architecture that integrates computer vision and deep learning techniques to continuously monitor the driver's condition. The system begins by capturing live video frames through a webcam, which serve as the input for further processing. Each frame is processed using OpenCV, while MediaPipe Face Mesh detects the driver's face and extracts facial landmarks required for behavioral analysis.

The extracted facial landmarks are simultaneously processed by the Eye Detection, Yawn Detection, and Head Position Detection modules. The Eye Detection module computes the Eye Aspect Ratio (EAR) to determine eye closure and identify drowsiness. The Yawn Detection module calculates the Mouth Aspect Ratio (MAR) to detect yawning, while the Head Position Detection module estimates the driver's head orientation to identify distractions caused by looking away from the road.

In parallel, the captured video frame is analyzed using the YOLOv8 object detection model to detect the presence and usage of a mobile phone while driving. The outputs from all detection modules are integrated by the Driver State Analysis module, which classifies the driver's condition as Alert, Drowsy, Yawning, or Distracted.

When an unsafe driving condition is detected, the system immediately generates visual warnings on the monitoring dashboard, activates a software-generated buzzer alert, captures a screenshot of the event, and records the event details with the current date and time in a CSV log file. The modular architecture enables efficient real-time processing while maintaining high accuracy and providing a scalable framework for future enhancements.

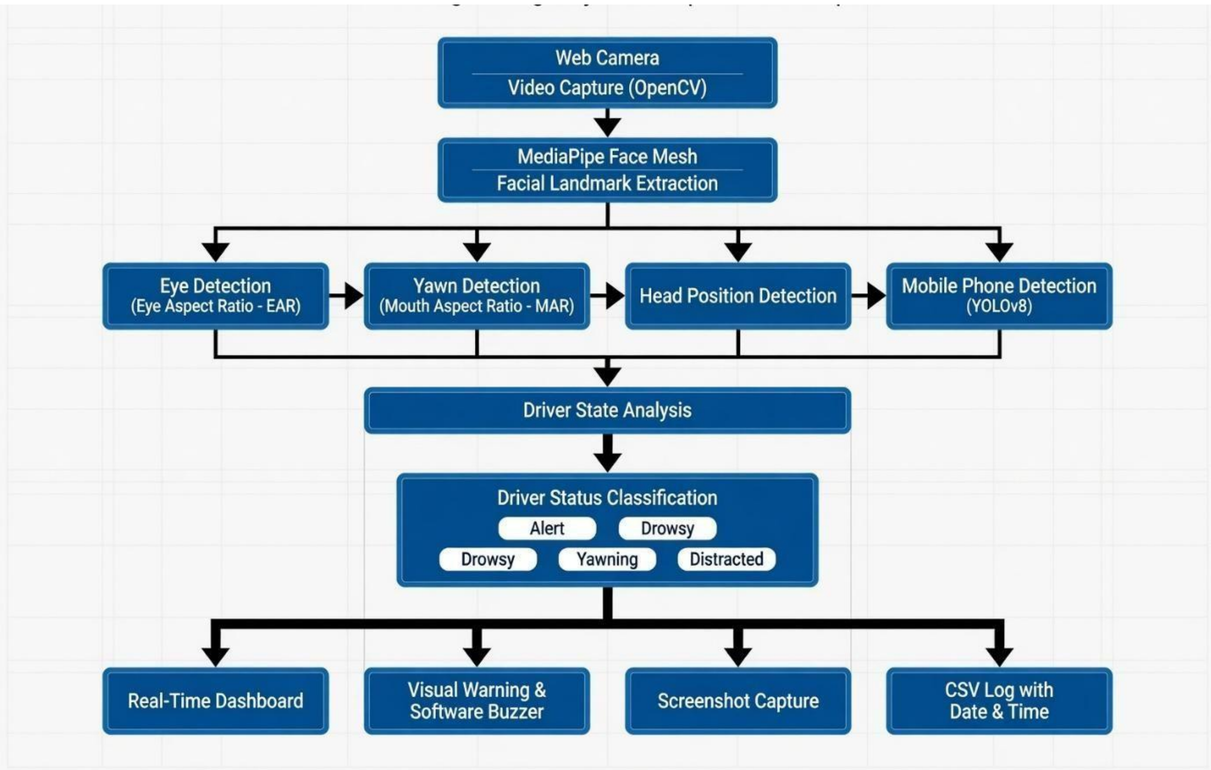

Figure 3.1: Overall System Architecture

3.3 Software and Hardware Requirements

The proposed Driver Drowsiness and Distraction Detection System is implemented using a combination of software libraries and standard computer hardware. The software environment provides the necessary tools for computer vision, deep learning, and real-time image processing, while the hardware components ensure smooth execution of the application.

3.3.1 Software Requirements

S. No.	Software	Purpose
1	Python 3.x	Programming language used for system development
2	Visual Studio Code	Integrated Development Environment (IDE)
3	OpenCV	Image processing and webcam video capture
4	MediaPipe Face Mesh	Facial landmark detection
5	YOLOv8 (Ultralytics)	Mobile phone detection using deep learning

6	NumPy	Numerical computations and array processing
7	CSV Module	Event logging and report generation
8	Datetime Module	Date and time recording for event logs
9	Winsound	Software-based buzzer alert (Windows)

Table 3.1 Software Requirements

3.3.2 Hardware Requirements

S. No.	Hardware	Specification
1	Laptop/Desktop	Intel Core i5 Processor or equivalent
2	RAM	Minimum 8 GB
3	Storage	Minimum 256 GB SSD
4	Webcam	HD USB or Built-in Camera
5	Operating System	Windows 10/11 (64-bit)
6	Internet Connection	Required only for installing libraries

Table 3.2 Hardware Requirements

3.4 System Modules

The proposed Driver Drowsiness and Distraction Detection System is developed using a modular architecture, where each module performs a specific task in detecting the driver's condition. These modules work together to continuously monitor the driver's behavior and determine whether the driver is alert or distracted.

3.4.1 Face Detection Module

The Face Detection Module is responsible for detecting the driver's face from the live video captured through the webcam. The system utilizes MediaPipe Face Mesh, which identifies 468 facial landmarks in real time with high accuracy. These landmarks provide the positional information required for subsequent analysis of the driver's facial features.

The detected facial landmarks form the basis for calculating the Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), and head orientation. Accurate face detection is essential because the performance of all subsequent modules depends on the precise localization of facial landmarks.

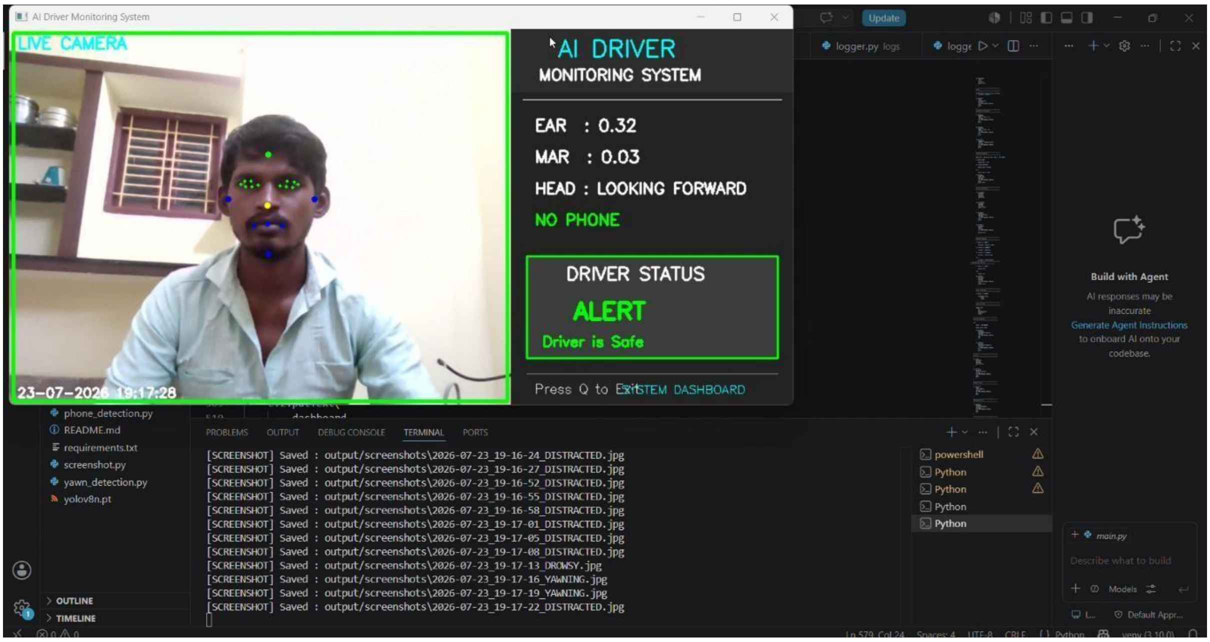

Figure 3.2: Face Detection using MediaPipe Face Mesh

3.4.2 Eye Detection Module

The Eye Detection Module continuously monitors the driver's eye movements to determine signs of drowsiness. Using the facial landmarks detected by MediaPipe Face Mesh, the system calculates the Eye Aspect Ratio (EAR), which measures the degree of eye openness.

When the EAR value remains below a predefined threshold for a continuous period, the system identifies the driver as drowsy. This method enables reliable fatigue detection without requiring additional sensors or wearable devices.

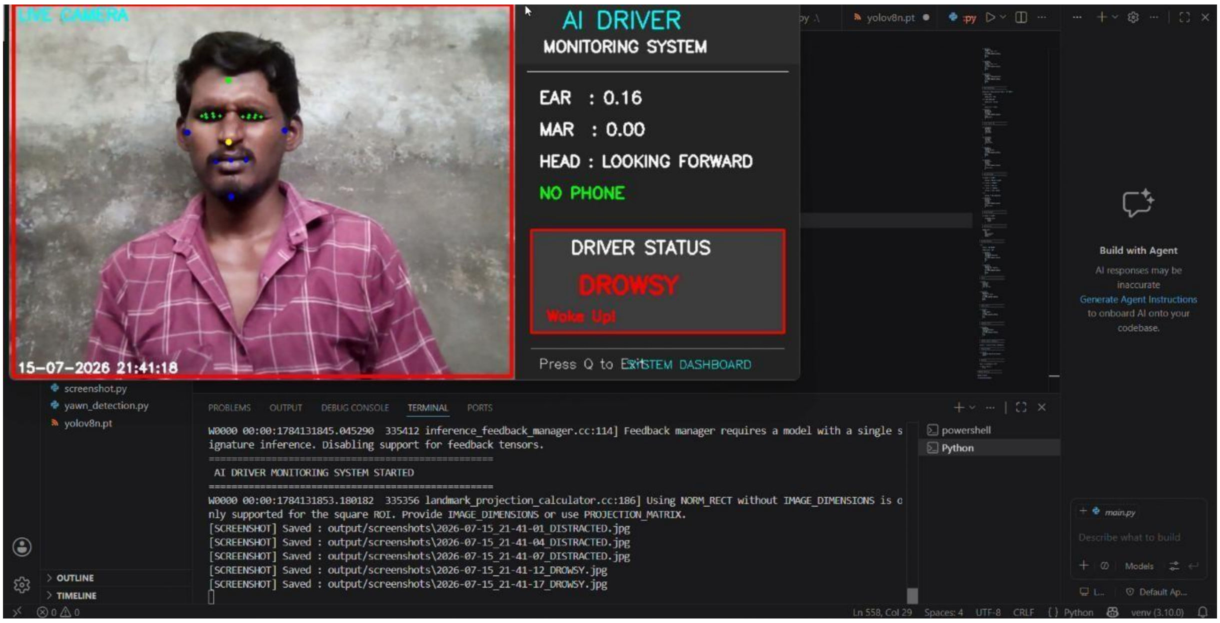

Figure 3.3: Eye Landmark Detection and EAR Calculation

3.4.3 Yawn Detection Module

The Yawn Detection Module identifies driver fatigue by analyzing mouth movements. The module computes the **Mouth Aspect Ratio (MAR)** using facial landmarks located around the lips. A significant increase in the MAR value indicates that the driver is yawning.

If the calculated MAR exceeds the predefined threshold, the system classifies the driver as yawning and generates an alert. This module enhances fatigue detection by complementing eye-based analysis.

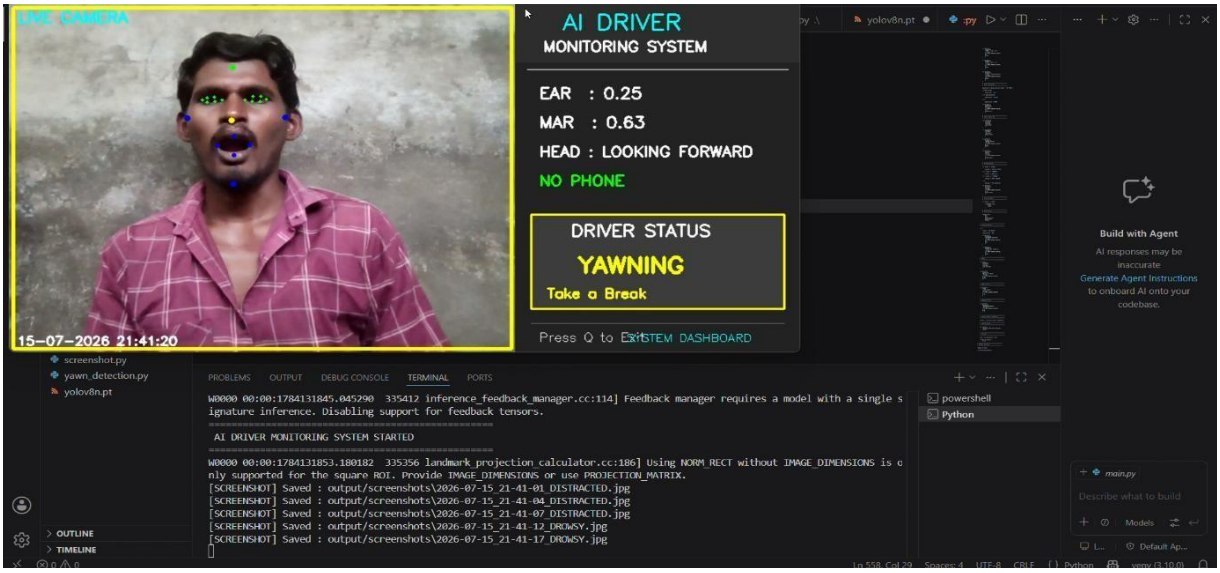

Figure 3.4: Mouth Landmark Detection for Yawning Analysis

3.4.4 Head Position Detection Module

The Head Position Detection Module determines the driver's head orientation using facial landmarks obtained from MediaPipe Face Mesh. The module analyzes the relative positions of facial features to estimate whether the driver is looking forward, left, right, up, or down.

If the driver's head remains away from the forward direction for a prolonged duration, the system identifies the behavior as driver distraction. This module improves safety by monitoring driver attention during vehicle operation.

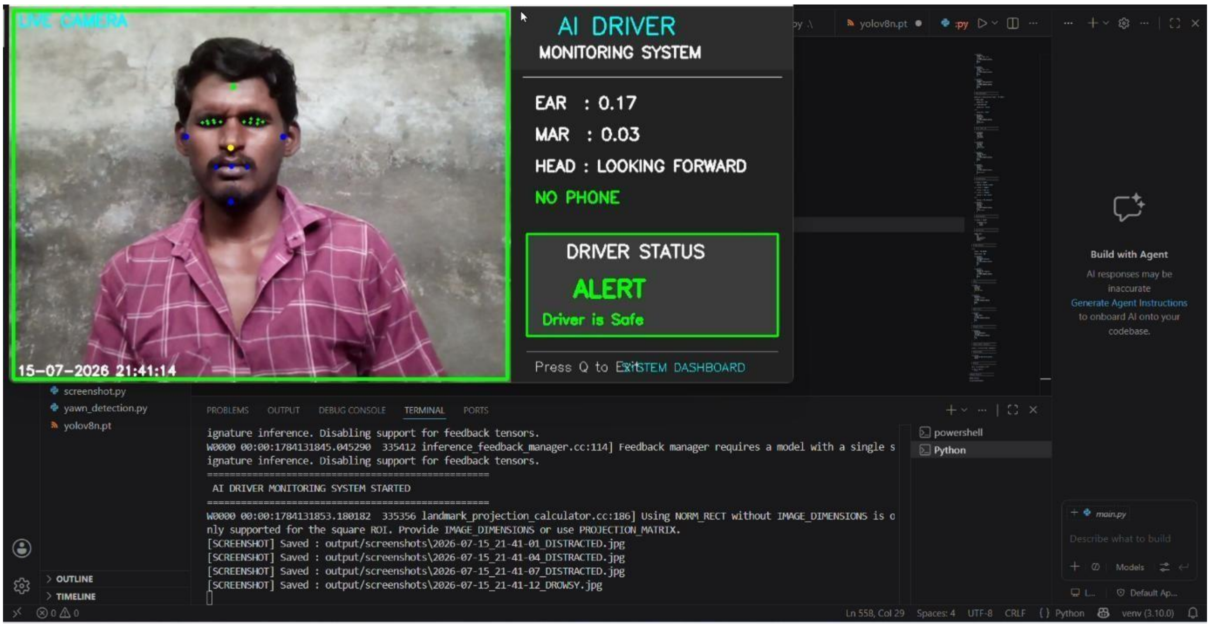

Figure 3.5.1: Driver Looking Forward

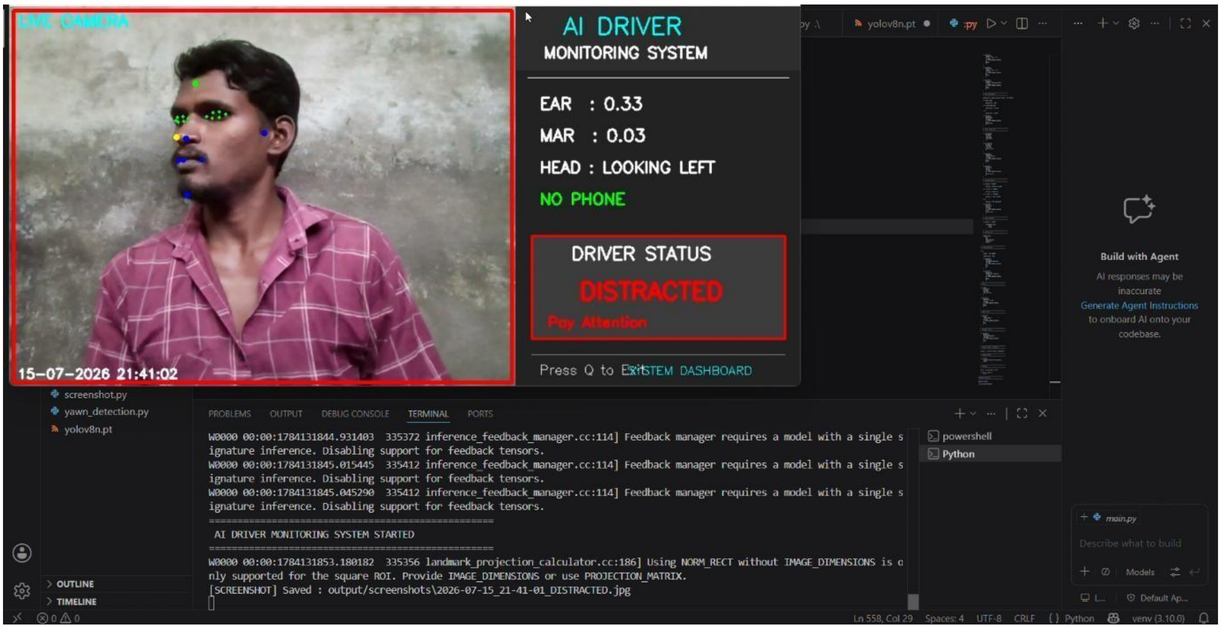

Figure 3.5.2: Driver Looking Left

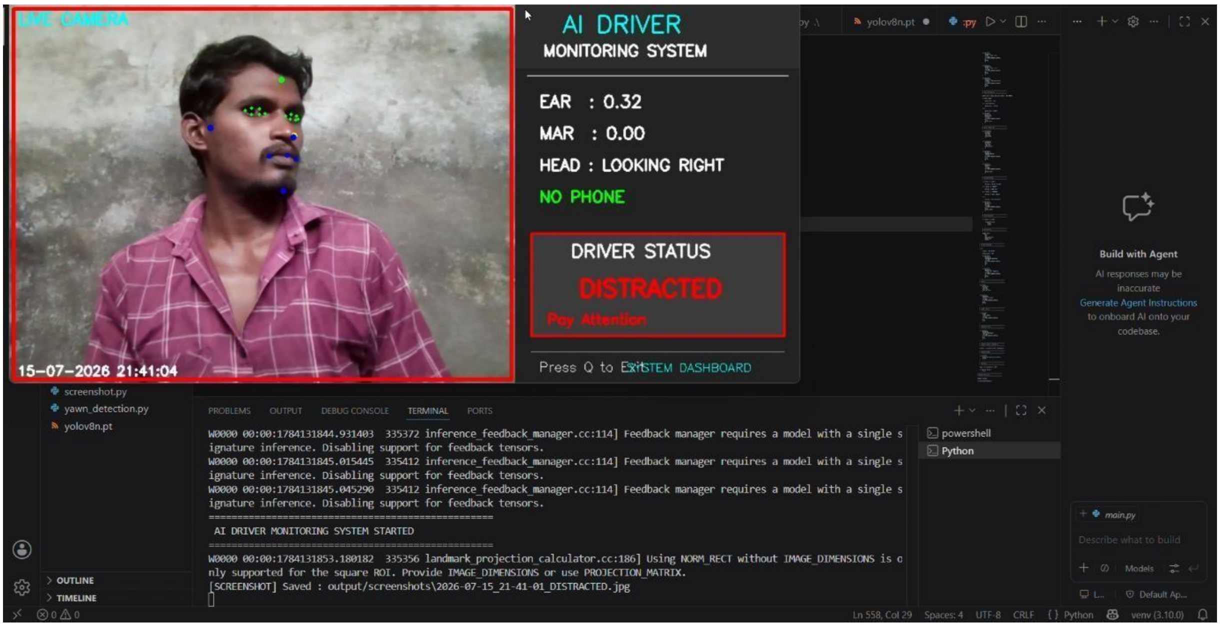

Figure 3.5.3: Driver Looking Right

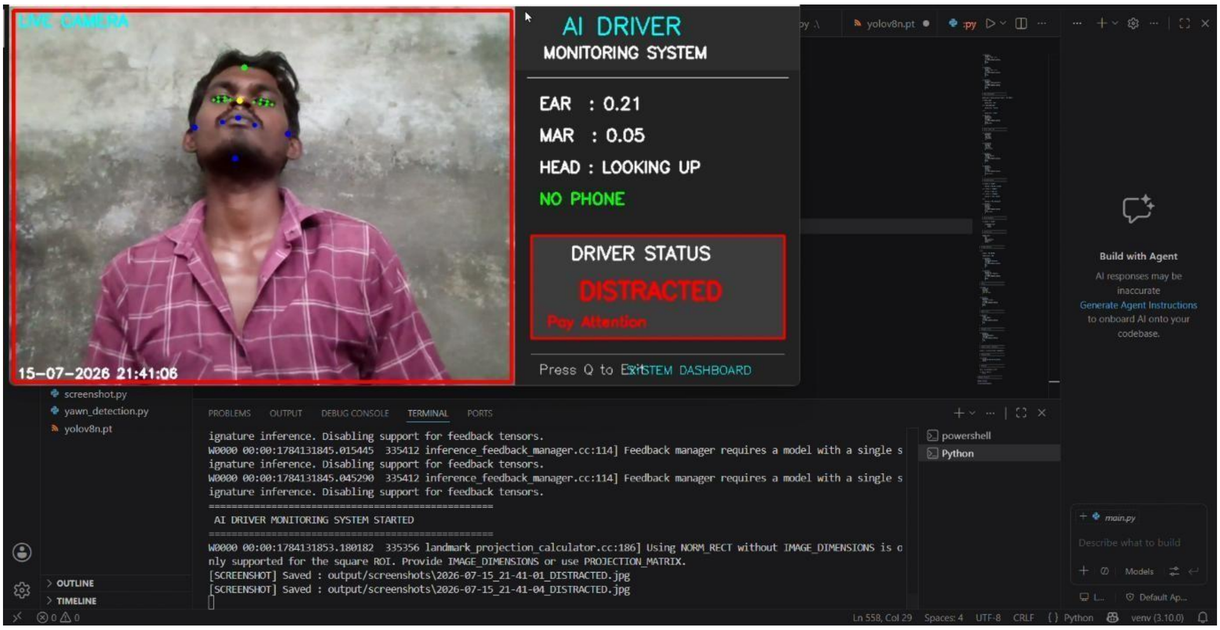

Figure 3.5.4: Driver Looking Up

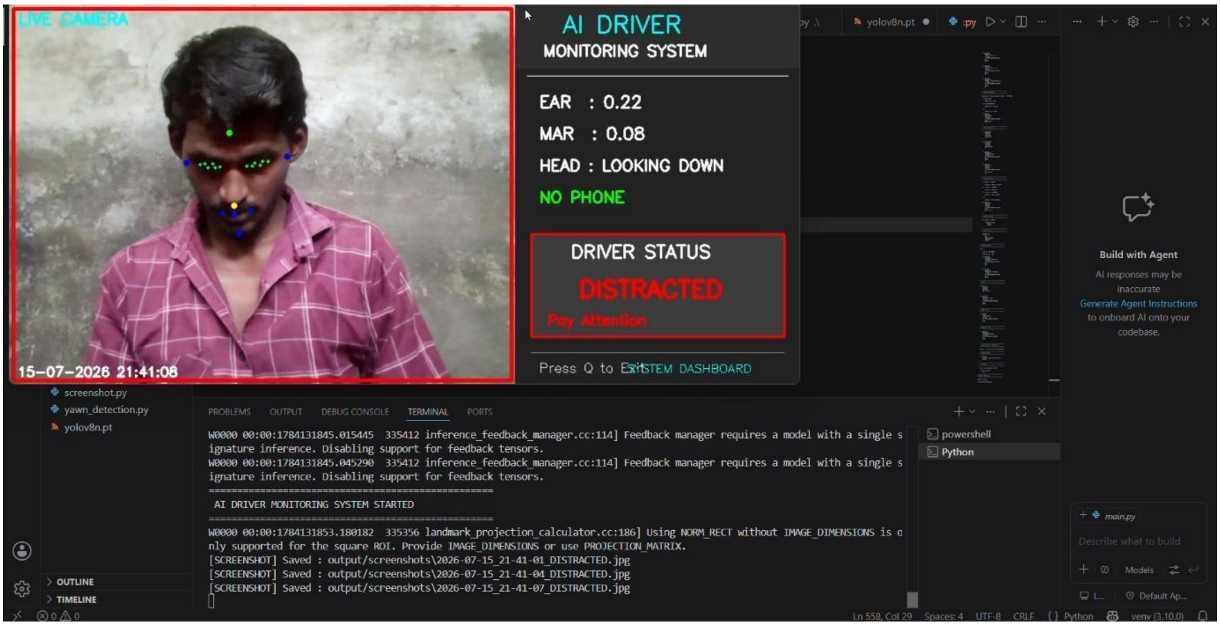

Figure 3.5.5: Driver Looking Down

3.4.5 Mobile Phone Detection Module

The Mobile Phone Detection Module employs the YOLOv8 deep learning object detection model to identify the presence of mobile phones in real time. The webcam frames are processed using YOLOv8, which detects mobile phones by drawing bounding boxes around the identified objects along with their confidence scores.

When the system detects that a mobile phone is being used while driving, it classifies the driver's state as distracted and generates an immediate warning. The use of YOLOv8 ensures fast and accurate object detection suitable for real-time applications.

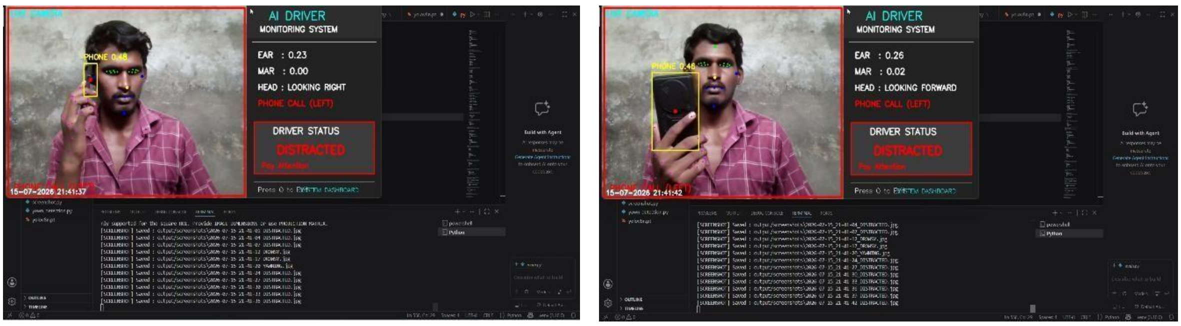

Figure 3.6: Mobile Phone Detection using YOLOv8

3.4.6 Driver Monitoring Dashboard

The Driver Monitoring Dashboard provides a real-time graphical interface displaying the driver's current condition. The dashboard continuously updates parameters such as Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), Head Position, Phone Detection Status, and the overall Driver Status.

Depending on the driver's behavior, the dashboard displays one of four conditions: Alert, Drowsy, Yawning, or Distracted. Color-coded warning messages enable quick identification of unsafe driving situations.

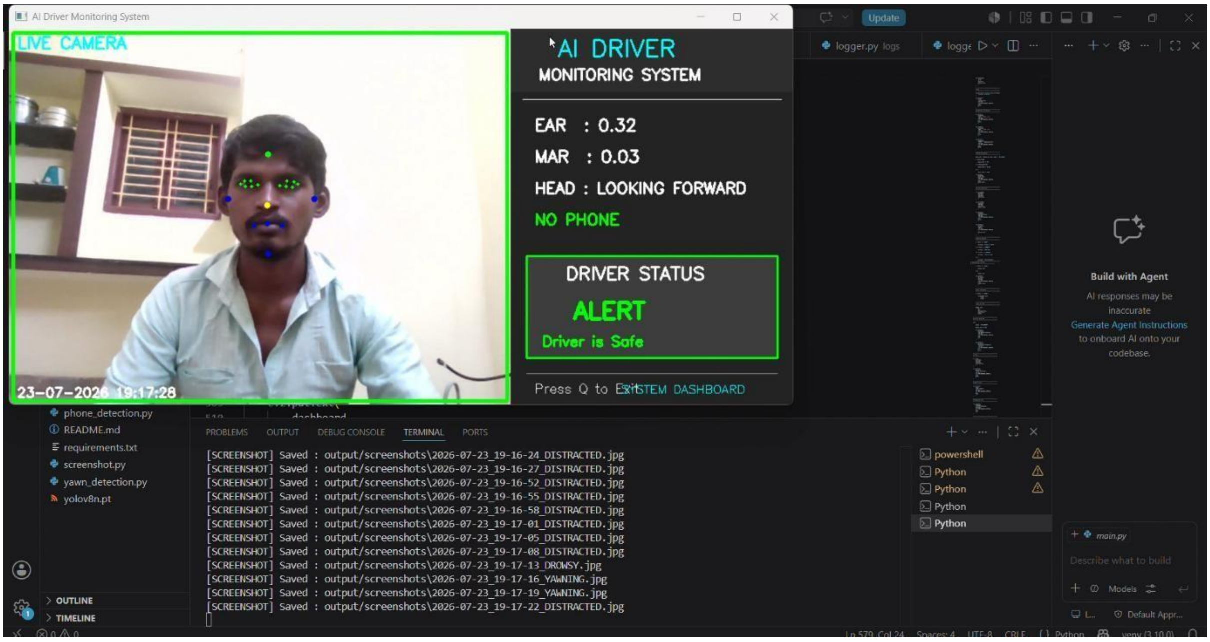

Figure 3.7: Driver Monitoring Dashboard

3.4.7 Screenshot and Logging Module

The Screenshot and Logging Module automatically records unsafe driving events for future analysis. Whenever the system detects drowsiness, yawning, distraction, or mobile phone usage, it captures a screenshot of the current video frame and stores it in a designated folder.

Additionally, the system maintains a CSV log file that records important information including the date, time, EAR value, MAR value, head position, phone detection status, and driver status. This feature provides a structured history of detected events and can be used for performance evaluation and incident analysis.

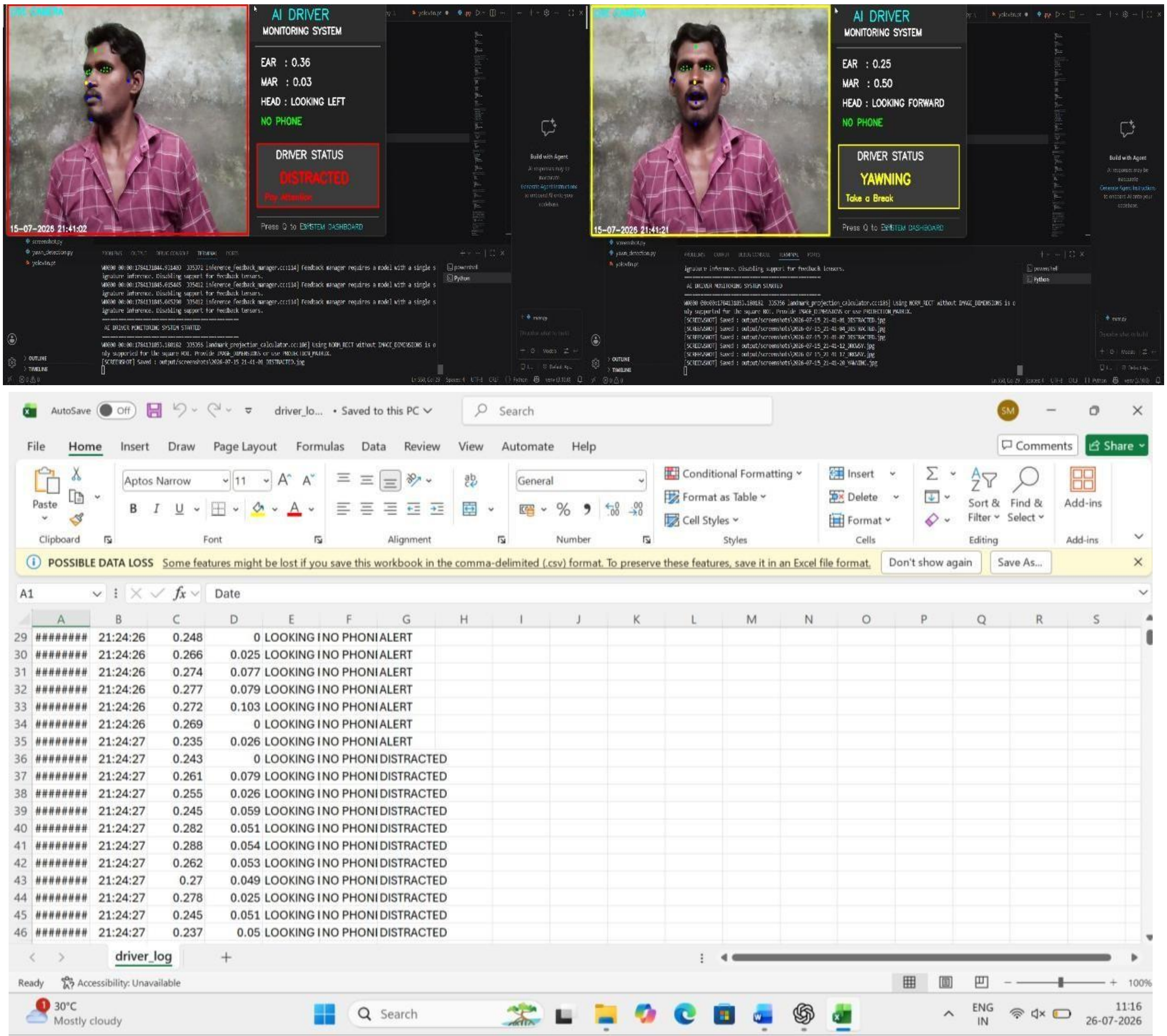

Figure 3.8: Event Screenshot and CSV Log File

CHAPTER 4

PERFORMANCE ANALYSIS AND VALIDATION

4.1 Testing Methodology

To evaluate the performance of the proposed Driver Drowsiness and Distraction Detection System using Computer Vision, a series of functional tests were conducted under different operating conditions using the developed software application. The primary objective of the testing process was to verify the correct operation of each detection module individually as well as the integrated system in a real-time environment.

The system was executed on a laptop using the built-in HD webcam to capture live video of the driver. During testing, different driver behaviors such as normal driving, prolonged eye closure, yawning, head movements, and mobile phone usage were intentionally performed to validate the functionality of the corresponding detection modules. The software continuously analyzed the captured video frames using OpenCV, MediaPipe Face Mesh, and the YOLOv8 object detection model, and displayed the detected driver status in real time.

Each module was evaluated based on its ability to correctly identify the intended driver behavior and generate the appropriate response. Whenever an unsafe condition was detected, the software successfully displayed warning messages, activated the software-generated buzzer alert, automatically captured screenshots, and recorded the event details along with the current date and time in the CSV log file. The testing process confirmed that all modules functioned together as an integrated real-time driver monitoring system.

Parameter	Specification
Processor	Intel Core i5 (or equivalent)
RAM	8 GB
Operating System	Windows 11 (64-bit)
Programming Language	Python 3.x
Development Environment	Visual Studio Code
Camera	Built-in HD Webcam (Tested) / External HD USB Webcam (Compatible)
Computer Vision Library	OpenCV
Facial Landmark Detection	MediaPipe Face Mesh
Object Detection Model	YOLOv8
Event Storage	CSV Log File and Screenshot Folder

Table 4.1 Experimental Setup

4.2 Functional Validation

The developed Driver Drowsiness and Distraction Detection System was validated by performing a series of real-time experiments under different driver conditions. Each functional module was tested individually and then evaluated as part of the integrated system to verify its performance. The expected outputs were compared with the actual outputs generated by the software. The results confirmed that all implemented modules operated correctly and produced the intended responses during testing.

The validation process involved observing different driver behaviors, including normal driving, prolonged eye closure, yawning, head movements, and mobile phone usage. Whenever the system detected an unsafe driving condition, it successfully generated the corresponding warning message, activated the software-generated buzzer alert, captured a screenshot of the event, and recorded the event details in the CSV log file. The successful execution of these functions demonstrates the reliability and effectiveness of the proposed system in continuously monitoring driver behavior.

S. No.	Function Tested	Expected Output	Actual Output	Result
1	Face Detection	Driver face detected	Face detected successfully	Pass
2	Eye Detection	EAR calculated	EAR calculated correctly	Pass
3	Drowsiness Detection	Detect prolonged eye closure	Drowsiness detected	Pass
4	Yawn Detection	Detect yawning	Yawning detected	Pass
5	Head Position Detection	Detect head orientation	Head position identified correctly	Pass
6	Mobile Phone Detection	Detect mobile phone usage	Phone detected using YOLOv8	Pass
7	Driver Status Classification	Display driver condition	Alert/Drowsy/Yawning/Distracted displayed correctly	Pass
8	Software Buzzer	Generate warning sound	Warning sound activated	Pass

9	Screenshot Capture	Save event image	Screenshot saved successfully	Pass
10	CSV Logging	Record event details	Data stored with date and time	Pass

Table 4.2 Functional Validation Results

4.3 Performance Evaluation

The performance of the proposed Driver Drowsiness and Distraction Detection System using Computer Vision was evaluated based on its real-time monitoring capability, detection accuracy, system responsiveness, and successful integration of multiple detection modules. During testing, the software continuously processed live video frames captured through the webcam and successfully analyzed the driver's facial features without noticeable interruption.

The MediaPipe Face Mesh framework provided stable facial landmark detection, enabling accurate computation of the Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), and head orientation. The YOLOv8 object detection model effectively identified mobile phone usage in real time and worked efficiently alongside the facial analysis modules. The integration of these techniques allowed the system to classify the driver's condition as Alert, Drowsy, Yawning, or Distracted with a prompt response.

The developed software also demonstrated reliable event management by automatically generating visual warnings, activating a software-generated buzzer alert, capturing screenshots of unsafe driving conditions, and recording event details with the current date and time in a CSV log file. Throughout the testing process, the system maintained stable operation and successfully executed all implemented functionalities, making it suitable for real-time driver monitoring applications.

Performance Parameter	Observation
Face Detection	Stable and consistent

Facial Landmark Detection	Accurate using MediaPipe Face Mesh
Eye Drowsiness Detection	Successfully detected prolonged eye closure
Yawn Detection	Correctly identified yawning events
Head Position Detection	Detected head orientation accurately
Mobile Phone Detection	Successfully detected mobile phone usage using YOLOv8
Real-Time Processing	Smooth execution without noticeable lag
Warning Generation	Visual warning and buzzer activated correctly
Screenshot Capture	Automatic image capture completed successfully
CSV Event Logging	Event details recorded with date and time
Overall System Performance	Satisfactory for real-time monitoring

Table 4.3 Performance Evaluation

4.4 Observations and Discussion

The experimental evaluation demonstrated that the proposed Driver Drowsiness and Distraction Detection System is capable of continuously monitoring driver behavior in real time using computer vision techniques. During testing, the system successfully detected facial landmarks using MediaPipe Face Mesh and accurately calculated the Eye Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR) for drowsiness and yawning detection. The head position estimation module effectively identified the driver's viewing direction, while the YOLOv8 model accurately detected the presence of a mobile phone within the camera frame.

The integration of multiple detection modules significantly improved the overall monitoring capability of the system. Instead of relying on a single parameter, the proposed software analyzed multiple behavioral cues simultaneously to determine the driver's condition. This multi-feature approach reduced the possibility of missed detections and provided more reliable driver state classification.

The automatic screenshot capture and CSV event logging functions operated successfully whenever an unsafe driving condition was detected. These features provide useful evidence for post-event analysis and system evaluation. Throughout the testing process, the software maintained stable performance without requiring additional hardware apart from a standard HD webcam, making the system cost-effective and easy to deploy.

Although the proposed system performed effectively under normal indoor lighting conditions, certain environmental factors such as poor illumination, partial facial occlusion, rapid head movements, and improper camera positioning may affect detection performance. Nevertheless, the developed software demonstrated satisfactory performance for real-time driver monitoring applications and validated the effectiveness of combining computer vision and deep learning techniques in a single integrated system.

Observation	**Result**
Real-time face detection	Successful
Eye drowsiness detection	Accurate under normal lighting
Yawn detection	Successfully identified yawning events
Head position estimation	Responsive to driver movements
Mobile phone detection	Reliable using YOLOv8
Warning generation	Triggered immediately after detection
Screenshot capture	Saved automatically during unsafe events
CSV logging	Correctly recorded event details with date and time

Overall software performance	Stable and satisfactory

Table 4.4 Key Observations

4.5 Summary

This chapter presented the performance analysis and validation of the proposed Driver Drowsiness and Distraction Detection System using Computer Vision. The developed software was successfully tested under various driver conditions to evaluate its real-time monitoring capabilities and overall functionality. The experimental results confirmed that the system accurately detected driver drowsiness, yawning, head movements, and mobile phone usage using computer vision and deep learning techniques.

The functional validation demonstrated that all implemented modules operated as expected, while the performance evaluation indicated stable and reliable execution during real-time operation. The automatic warning generation, screenshot capture, and CSV event logging features further enhanced the effectiveness of the system by providing immediate alerts and maintaining records of unsafe driving events.

CHAPTER 5
LIMITATIONS AND FUTURE SCOPE

5.1 Introduction

The proposed Driver Drowsiness and Distraction Detection System using Computer Vision successfully demonstrated the ability to monitor driver behavior in real time by detecting drowsiness, yawning, head movements, and mobile phone usage. The experimental results confirmed that the developed software operates effectively under normal testing conditions using a standard HD webcam and computer vision techniques.

Despite its successful performance, the current implementation has certain limitations that may affect its operation under specific conditions. Identifying these limitations is important for understanding the scope of the present work and for guiding future improvements. This chapter discusses the existing limitations of the proposed system and highlights potential enhancements that can further improve its accuracy, reliability, and practical applicability in real-world driver monitoring systems.

5.2 Limitations

Although the proposed Driver Drowsiness and Distraction Detection System using Computer Vision demonstrated satisfactory performance during testing, certain limitations were observed that may affect its performance under different operating conditions.

The system relies on continuous facial visibility for accurate detection. If the driver's face is partially covered by accessories such as sunglasses, masks, or hats, the accuracy of facial landmark detection may decrease, affecting the computation of the Eye Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR).

The performance of the system is influenced by environmental lighting conditions. Low illumination, excessive brightness, or shadows may reduce the quality of the captured video, thereby affecting both facial landmark detection and mobile phone detection.

The mobile phone detection module identifies phones only when they are clearly visible within the camera's field of view. If the phone is completely hidden, partially occluded, or outside the camera frame, the detection accuracy may be reduced.

Rapid head movements, extreme face rotations, or improper camera positioning may occasionally affect the accuracy of head position estimation and facial landmark tracking. Maintaining an appropriate camera angle and distance helps improve the overall performance of the system.

The current implementation has been evaluated primarily with a single driver in controlled indoor environments. Additional testing under diverse real-world driving conditions, including different lighting environments and multiple users, would further improve the robustness and generalizability of the system. Despite these limitations, the developed software successfully demonstrates the effectiveness of integrating computer vision and deep learning techniques for real-time driver drowsiness and distraction detection.

5.3 Future Scope

The proposed Driver Drowsiness and Distraction Detection System using Computer Vision provides a strong foundation for intelligent driver monitoring and can be further enhanced through several improvements. Future developments can focus on increasing the system's accuracy, robustness, and applicability in real-world driving environments.

The current software can be deployed on embedded computing platforms such as Raspberry Pi or NVIDIA Jetson to enable a compact and portable driver monitoring solution suitable for installation inside vehicles. This would improve the practicality of the system for real-time deployment.

The performance of the system can be enhanced by training advanced deep learning models on larger and more diverse datasets collected under different lighting conditions, weather conditions, and driver behaviors. Such improvements would increase the robustness and reliability of the detection process.

Additional driver monitoring features such as seat belt detection, smoking detection, eating or drinking detection, and driver identity recognition can be incorporated to provide a more comprehensive assessment of driver behavior and safety.

The system can also be integrated with Advanced Driver Assistance Systems (ADAS) to enable automatic safety responses, such as warning notifications or assistance mechanisms, when unsafe driving behavior is detected. Furthermore, cloud connectivity and Internet of Things (IoT) technologies can be incorporated to support remote monitoring, centralized data storage, and long-term analysis of driver behavior.

CHAPTER 6

APPENDICES

APPENDIX – 1

PSUEDOCODE

A.1 Face Detection Module

```
BEGIN
Capture video frame
Convert image to RGB
Process frame using MediaPipe Face Mesh
IF face landmarks detected THEN
    Extract facial landmarks
ELSE
    Display "No Driver Detected"
END IF
END
```

A.2 Eye Detection Module (EAR Calculation)

```
BEGIN
Obtain eye landmark coordinates
Calculate vertical eye distances
Calculate horizontal eye distance
Compute Eye Aspect Ratio (EAR)
IF EAR < Threshold THEN
    Driver State ← Drowsy
ELSE
    Driver State ← Alert
END IF
END
```

A.3 Yawn Detection Module (MAR Calculation)

```
BEGIN
Obtain mouth landmark coordinates
Calculate mouth opening
Compute Mouth Aspect Ratio (MAR)
IF MAR > Threshold THEN
    Driver State ← Yawning
END IF
END
```

A.4 Head Position Detection Module

```
BEGIN
Read facial landmark coordinates
Estimate head orientation
IF Head facing Forward THEN
    Head Status ← Looking Forward
ELSE
    Head Status ← Distracted
END IF
END
```

A.5 Mobile Phone Detection Module (YOLOv8)

```
BEGIN
Load YOLOv8 model
Read current video frame
Detect objects
IF Mobile Phone detected THEN
    Phone Status ← Detected
ELSE
```

```
    Phone Status ← Not Detected
END IF
END
```

A.6 Driver Status Classification

```
BEGIN
Read EAR
Read MAR
Read Head Position
Read Phone Detection
IF EAR < Threshold THEN
    Status ← Drowsy
ELSE IF MAR > Threshold THEN
    Status ← Yawning
ELSE IF Phone Detected THEN
    Status ← Distracted
ELSE IF Head Position ≠ Looking Forward THEN
    Status ← Distracted
ELSE
    Status ← Alert
END IF
Display Status
END
```

A.7 Event Logging Module

```
BEGIN
Read current date and time
Collect EAR
Collect MAR
Collect Head Position
Collect Phone Status
Collect Driver Status
Store all values in CSV file
END
```

A.9 Main Program

```
BEGIN
Initialize webcam (If Has)
Load all detection modules
WHILE webcam is active DO
    Capture video frame
    Perform Face Detection
    Perform Eye Detection
    Perform Yawn Detection
    Perform Head Position Detection
    Perform Phone Detection
    Classify Driver Status
    Save Screenshot if required
    Save Event Log
    Display Monitoring Screen
END WHILE
END
```

APPENDIX B

PROJECT FOLDER STRUCTURE

The proposed Driver Drowsiness and Distraction Detection System using Computer Vision follows a modular folder structure to improve code organization, readability, and maintenance. Each Python module performs a dedicated function, while separate folders are used to store screenshots, logs, and AI model files.

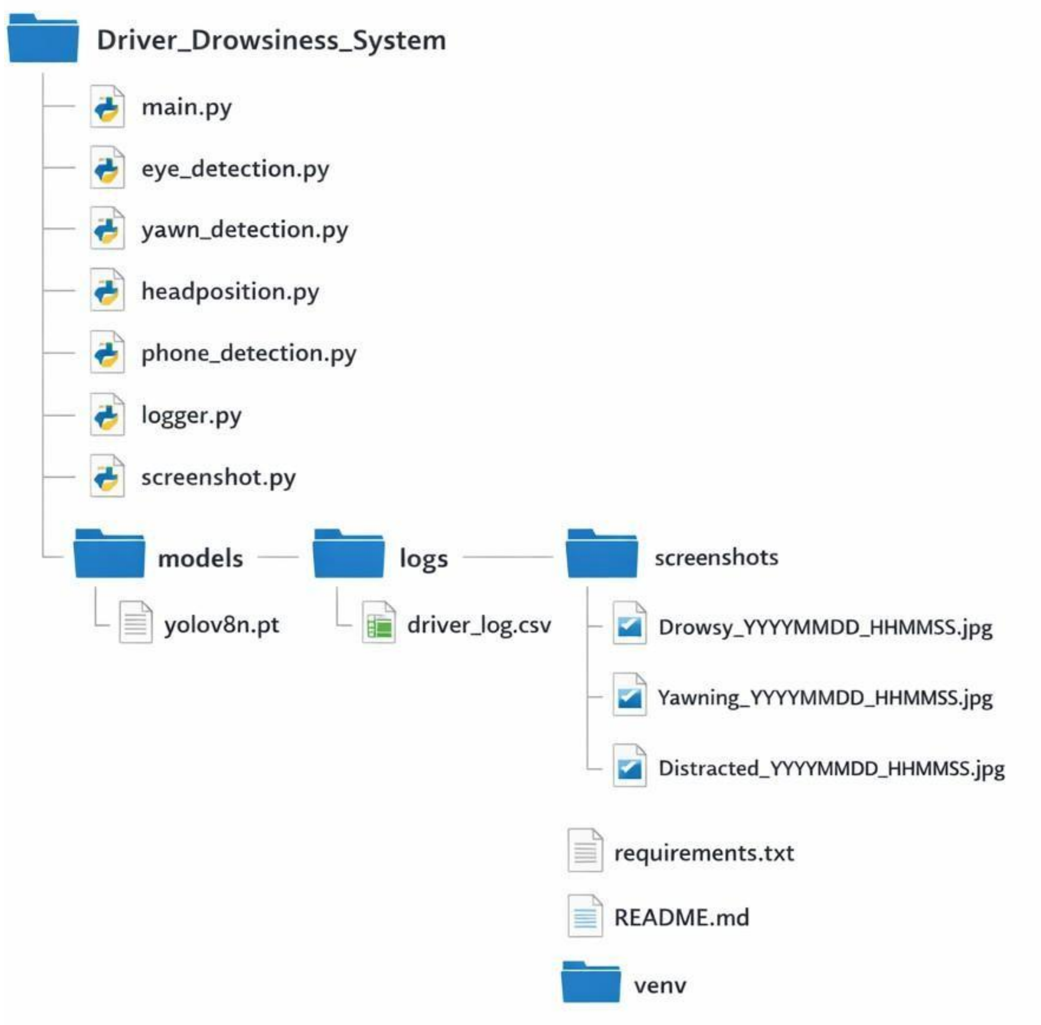

Figure C.1: Project Folder Structure of the Driver Drowsiness and Distraction Detection System using Computer Vision.

File / Folder	Description
main.py	Main program that initializes the system, processes webcam frames, and integrates all detection modules.
eye_detection.py	Detects eye landmarks and calculates the Eye Aspect Ratio (EAR) for drowsiness detection.
yawn_detection.py	Detects mouth landmarks and calculates the Mouth Aspect Ratio (MAR) for yawn detection.
headposition.py	Estimates the driver's head orientation using facial landmarks.
phone_detection.py	Uses the YOLOv8 model to detect mobile phone usage while driving.
logger.py	Records driver monitoring events with the current date and time in a CSV log file.
screenshot.py	Captures and stores screenshots whenever an unsafe driving condition is detected.
models/	Contains the pretrained YOLOv8 model used for object detection.
logs/	Stores the generated CSV log file containing driver monitoring records.
screenshots/	Stores screenshots captured during drowsiness, yawning, or distraction events.
requirements.txt	Lists the Python libraries and dependencies required to execute the project.
README.md	Provides project overview and execution instructions.
venv/	Contains the Python virtual environment and installed packages.

Table C.1 Description of Project Files and Folders

CHAPTER 7
CONCLUSSION

The Driver Drowsiness and Distraction Detection System using Computer Vision was successfully designed and developed as a real-time software solution for monitoring driver alertness and detecting unsafe driving behaviors. The system integrates OpenCV, MediaPipe Face Mesh, and the YOLOv8 object detection model to analyze live video captured through an HD webcam. By calculating the Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), head position, and detecting mobile phone usage, the system effectively identifies drowsiness, yawning, and driver distraction. Whenever an unsafe condition is detected, the system immediately generates visual warnings, activates a software-based buzzer alert, captures event screenshots, and records the event details with the current date and time in a CSV log file for future reference. The modular implementation using Python provides an efficient, reliable, and easily maintainable solution for real-time driver monitoring. Experimental testing demonstrated that the proposed system performs effectively under normal operating conditions and accurately identifies unsafe driving behaviors, thereby contributing to improved road safety. Overall, the project successfully achieved its objectives and demonstrates the practical application of computer vision and artificial intelligence in developing an intelligent, low-cost, non-contact driver monitoring system with the potential for future integration into advanced vehicle safety technologies.

CHAPTER 8

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